School of Science, Computing and Emerging Technologies

Unit Outline

COS40003

Concurrent Programming

Semester September 2026

Please read this Unit Outline carefully. It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information





"Swinburne University of Technology recognises the historical and cultural significance of Australia's Indigenous history and the role it plays in contemporary education

Each day in Australia, we all walk on traditional Indigenous land

We therefore acknowledge the traditional custodians of the land that our Australian campuses currently occupy, the Wurundjeri people, and pay respect to Elders past and present, including those from other areas who now reside on Wurundjeri land"

PART A: Unit Summary

Unit Code(s)		COS40003
Unit Title		Concurrent Programming
Duration		One semester or equivalent
Total Contact Hours		36 hours
Requisites:		
	Pre-requisites	COS20007 Object Oriented Programming OR SWE20004 Technical Software Development OR COS30008 Data Structures and Patterns OR COS30043 Interface Design and Development
	Co-requisites	Nil
	Concurrent pre-requisites	Nil
	Anti-requisites	COS30003 Advanced .NET Programming
	Assumed knowledge	Nil
Credit Points		12.5 credit points
Campus/Location		Swinburne Vietnam
Mode of Delivery		Blended
Assessment Summary		Report 1 (Individual) 30% Question Test (Individual) 40% Programming Tests (Individual) 20% Report 2 (Individual) 10%

Aims

This unit of study aims to introduce students to the design and development of concurrent and parallel programs.

Unit Learning Outcomes

Students who successfully complete this Unit should be able to:

1 Design and develop complex concurrency control utilities using threading and primitive synchronisation mechanisms of the language used

2 Apply contemporary programming languages and frameworks to design and develop concurrent and parallel programs

3 Analyse algorithm and utility designs to identify, explain, and correct issues related to safety, liveness, performance and reusability in concurrent contexts

4 Compare and contrast concurrency support provided by different programming language frameworks

Graduate Attributes

This unit may contribute to the development of the following Swinburne Graduate Attributes:

- Communication skills
- Teamwork skills
- Digital literacies

Content

- Concurrency mechanisms
- Concurrency control primitives
- Concurrency control utilities
- Design forces related to concurrent software development

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- Extension and late submission rules updated
- Assessments have been re-evaluated.

Unit Teaching Staff

Name	Role	Location	Email	Consultation Times
Mr. Nguyen Van Cong (Dennis)	Unit Coordinator	Hanoi	vancongnguyen@swin.edu.au	By email appointment
Mr. Pham Ky Trung	Lead Coordinator	HCM City	ktpham@swin.edu.au	By email appointment

Learning and Teaching Structure

Activity	Total Hours	Hours per Week	Teaching Period Weeks
Lectures	24 hours	2 hours	Weeks 1 to 12
Labs	12 hours	1 hour	Weeks 1 to 12
Independent Learning	114 hours	9.5 hours	Weeks 1 to 12

Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	14 Sep	Introduction	
2	21 Sep	Process	
3	28 Sep	Scheduling	
4	5 Oct	Thread	
5	12 Oct	Lock (I)	
6	19 Oct	Lock (II)	Report I due, Oct 25, 23:59
7	26 Oct	Condition Variable	no labs on AFL Grand Final
8	2 Nov	Semaphore	
9	9 Nov	Concurrency Bugs	
10	16 Nov	Question Test	Online test – during Week 10 lecture time
11	23 Nov	Other Topics or Guest Lecture (I)	
12	30 Nov		Other Topics or Guest Lecture (II) Programming Test, Week 12
13			Report II due, Sunday Dec 6, 23:59.

Assessment

a) Assessment Overview

Tasks and Details	Individual or Group	Weighting	Unit Learning Outcomes that this assessment task relates to	Assessment Due Date
1. Report I	Individual	30%	1, 3, 4	End of week 6
2. Question Test	Individual	40%	1, 2, 3, 4	Week 10 lecture
3. Programming Test	Individual	20%	1, 2, 3, 4	Week 12
4. Report II	Individual	10%	1, 2, 3, 4	End of Week 12

b) Minimum requirements to pass this Unit

To pass this unit, you must:

- achieve an overall mark for the unit of 50% or more, and

c) Examinations

If the unit you are enrolled in has an official examination, you will be expected to be available for the entire examination period including any Special Exam period.

d) Submission Requirements

Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin plagiarism checking service.

Please ensure you keep a copy of all assessments that are submitted.

In cases where a hard copy submission is required an Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Current Students web site (see Part C).

In this unit:

Reports and Test code need to be submitted online through the Canvas assessment submission system which integrates with the Turnitin plagiarism checking service.

Please ensure you keep a copy of all assessments that are submitted.

e) Extensions and Late Submission

Late Submissions - Unless an extension has been approved, late submissions will not be accepted or result in a penalty.

For Report I and Report II, you will be penalised 5% of total assessment value for each working day the task is late, up to a maximum of 7 working days. After 7 working days, a zero result will be recorded.

For Question Test and Programming Test, late submissions are not accepted.

f) Referencing

To avoid plagiarism, you are required to provide a reference whenever you include information from other sources in your work. Further details regarding plagiarism are available in Section C of this document.

Referencing conventions required for this unit are: any of your choice. Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/

Recommended Reading Materials

The Library has a large collection of resource materials, both texts and current journals. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

- Remzi Arpaci-Dusseau,Andrea Arpaci-Dusseau: Operating Systems: Three Easy Pieces. Free Online Textbook. http://pages.cs.wisc.edu/~remzi/OSTEP/

- Brian Göetz, Tim Peierls, Joshua Bloch, Joseph Bowbeer, David Holmes, Doug Lea: Java Concurrency in Practice. Addition-Wesley Professional. 2005.

PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage http://www.swinburne.edu.au/student/

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The Student Charter describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property. You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the Student Academic Misconduct Regulations, Student General Misconduct Regulations and the People, Culture and Integrity Policy. Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response

Canvas

You should regularly access the Swinburne learning management system, Canvas, which is available via the Current Students webpage or https://swinburne.instructure.com/ Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources; contributing fairly to group work; and completing tasks, tests and exams without cheating. Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarising, cheating and seeking an unfair advantage with regards to an exam or assessment are all breaches of academic integrity and treated as academic misconduct.

Plagiarism is submitting or presenting someone else's work as though it is your own without full and appropriate acknowledgement of their ideas and work. Examples include:

- using the whole or part of computer program written by another person as your own
- using the whole or part of somebody else's written work in an essay or other assessable work, including material from a book, journal, newspaper article, a website or database, a set of lecture notes, current or past student's work, or any other person's work
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another
- using realia created by another person, such as objects, artefacts, costumes, models • submitting assessments that have been developed by another person or service (paid or unpaid), often referred to as contract cheating
- presenting or submitting assignments or other work in conjunction with another person or group of people when that work should be your own independent work. This is regardless of whether or not it is with the knowledge or consent of the other person(s). Swinburne encourages students to talk to staff, fellow students and other people who may be able to contribute to a student's academic work but where an independent assignment is required, the work must be the student's own
- enabling others to plagiarise or cheat, including letting another student copy your work or by giving access to a draft or completed assignment

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to expulsion from the unit and, in the extreme, exclusion from Swinburne.

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration will be submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component.

Accessibility needs

Sometimes students with a disability, a mental health or medical condition or significant carer responsibilities require reasonable adjustments to enable full access to and participation in education. Your needs can be addressed by Swinburne's AccessAbility Services by negotiating and distributing an 'Education Access Plan'. The plan makes recommendations to university teaching and examination staff. You must notify AccessAbility Services of your disability or condition within one week after the commencement of your unit to allow the University to make reasonable adjustments.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

If you are not satisfied with the result of an assessment, you can ask the Unit Convenor to review the result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor will review your result to determine if your result is appropriate.

If you are dissatisfied with the outcomes of the review, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If you are dissatisfied with the outcome of the discussion or would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See https://www.swinburne.edu.au/corporate/feedback/

Should you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Academic Student Support Officer at Swinburne Student Life.

For an appointment, please call 19006412 or lodge a query via Student Portal at https://portal.swin.edu.vn